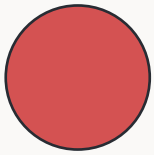
Learning from examples

Term 3 | Week 6 | Lesson reference

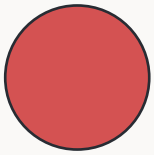
Teacher: print these cards or display them. These are labelled examples for a classroom analogy. No real computer model is being trained by sorting paper.

A1



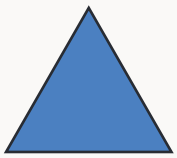
circle

A2



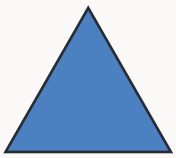
circle

A3



triangle

A4



triangle

Which shape is each example?

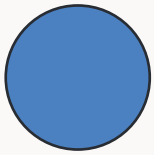
What do you notice about the colours?

Could a circle be blue?

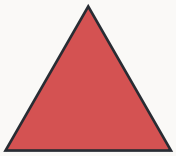
Check new examples

These test cards have no answer label. Make a guess, then check the shape. A person checks by looking at the shape, not just its colour.

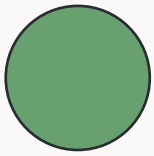
T1



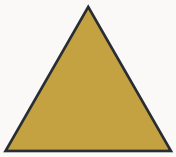
T2



T3



T4



Card	Our guess	Checked shape
T1		
T2		
T3		
T4		

More useful examples

Draw one circle in a new colour. Draw one triangle in a new colour. These examples help us see that colour and shape are different features.

My new circle

My new triangle

Talk to a partner

What mistake could someone make if they looked only at colour? How could we check?

Teacher notes and answers

Two lessons

Lesson 1: label A1-A4, discuss a possible rule and create varied examples. Lesson 2: test T1-T4, record predictions and check them. Pupils may infer the correct shape rule immediately; accept this and use a deliberate colour-only guess as the contrast.

Answers

A1 and A2 are circles; A3 and A4 are triangles. T1 and T3 are circles; T2 and T4 are triangles. A colour-only guess could mislabel the blue circle and red triangle. There can be more than one possible guess from a small example set; discuss the feature used.

Explain the analogy carefully

Actual machine-learning systems use numerical representations and learning procedures to find patterns. Our paper activity makes examples, labels and checks visible. It does not prove how a particular model works or guarantee that more examples always fix an error.

Optional teacher demonstration

If the school connection allows, preview Code.org AI for Oceans before class. Demonstrate only the initial fish/not-fish task on the teacher device. It gives a real model example; it is optional, and the paper lesson works without login or internet. Do not enter pupil details or upload pupil images.

Teacher references

Original SIAMIS activities informed by the sources below. These are adaptations, not reproduced publisher lesson plans. Links checked 21 October 2026.

Day of AI - Curriculum - Learning from Patterns listing
https://dayofai.org/curriculum-resources

Code.org - AI for Oceans
https://studio.code.org/courses/oceans/units/1/lessons/1